ECVOL-bench: An Identity-Controlled Evaluation Protocol and Audit Toolkit for Multimodal Earnings-Call Volatility Prediction

Anonymous ACL submission

Abstract

For about a decade, researchers have predicted how much a company's stock will move after its quarterly earnings call from what executives say and how they sound, and many report beating classical forecasts built from past price swings. Nearly all of those results are scored on one set of label and split files handed down from paper to paper. We audit those files and find that they cannot support the claim. They leave no gap between the training and test periods although every target looks 30 days ahead. Between 44% and 80% of their test calls come from companies the model also saw in training. And they place "day zero" on the day of the call, although most companies release their numbers before the market opens that morning. We release an evaluation protocol and an audit toolkit that make these problems measurable. The protocol recovers company names and dates for a public audio corpus that ships without them, times each release from SEC filings, provides targets under both horizon definitions in use, and splits that keep training and test apart in time and in company. The toolkit runs identity checks on any file of predictions. Under these checks, modern open text and audio models do not improve on a matched model that uses past volatility alone. Text embeddings mostly identify which company is talking (89.5% probe accuracy), and on a date-ordered split with unseen companies no text, audio or combined model reduces error against that matched model by a margin fixed in advance. Five published models rerun under the same checks show a similar pattern. The test error of their text models rises by 12–82% once test companies are unseen, while price-only models show no consistent change. Scoring the frozen pipeline on 475 calls from late 2025, after every model's training cutoff, changes none of this. Content signal may still exist in these calls. It has to be shown to survive these checks, and the released protocol, baselines and audit command make that the starting point for the next study.

1 Introduction

An earnings call is the one regular occasion on which a company's managers speak at length about their business. Since Qin and Yang (2019), a line of papers has used both the transcript and the audio of these calls to predict how volatile the stock will be over the following 3 to 30 trading days, and HTML (Yang et al., 2020), MAEC (Li et al., 2020), VolTAGE (Sawhney et al., 2020a), DialogueGAT (Sang and Bao, 2022), KeFVP (Niu et al., 2023) and ECC Analyzer (Cao et al., 2024) all report beating classical forecasts. Yu et al. (2025) gave a reason to look again. On a corpus of their own, transcript representations mostly told the model *which company* was speaking, and a forecast from the company's own past volatility beat every transcript model.

That diagnosis could not be measured on the files the field scores on. Nearly every paper in this line uses one set of label and split files, handed down from paper to paper. We audited them (§3). They leave no gap between training and test dates even though each target runs 30 days past its call. Most test calls come from companies that are also in training, and no paper reports what happens when they are not. They define horizons in calendar days, which later papers describe as trading days. They put day 0 on the date of the call, although most companies publish their results before the market opens that day. Their labels cannot be regenerated from prices under any single stated rule. The only open corpus with raw audio, FinCall-Surprise (Shu et al., 2026a), has no ticker, date or speaker fields, so the identity checks cannot be run on it at all. No published model had been rerun under such checks, and none had been scored on calls after its training cutoff. Figure 1 shows the two properties that matter most. This paper makes the diagnosis measurable on the files in use and releases the tools for it.

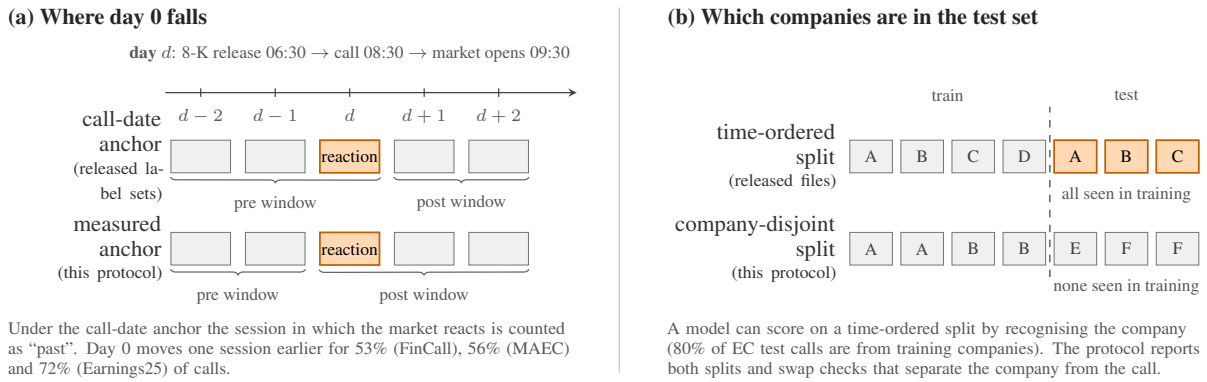


Figure 1: **Two properties of the shared test files.** (a) Most companies release results before the open, so anchoring day 0 on the call date leaves the session in which the market reacts inside the pre window. The protocol anchors on the measured release time. (b) A time-ordered split keeps the same companies on both sides of the boundary. The protocol adds a company-disjoint split and swap checks that make the difference measurable.

What we contribute. (1) **An evaluation protocol**, released as ECVOL-bench. We recover company identities and dates for FinCall-Surprise (92.9% of calls, and a 60-call manual audit passes 60/60) and measure release times from 8-K filings. The release ships three target families under both horizon definitions, splits with a 30-session gap and company-disjoint splits, checksums for every file, and a harness in which every table comes from one command (§3). It redistributes no raw audio or transcripts. What it adds to the existing corpora is the identity and timing tables, the targets, splits and derived features, and the harness. (2) **An audit** of the shared label and split files (Table 1). (3) **An audit toolkit.** Identity checks (ticker probes, same-company and random content swaps, company-disjoint evaluation) released as a command that works on *any* prediction file. (4) **Results** for modern open-weight text and audio models against a matched price-only model, with a margin fixed in advance (§5). (5) **A reproduction study** of the five runnable models in this line, on their own data where it is still available, with the same checks, plus a code-availability audit of seventeen papers. (6) **A post-cutoff test** of the frozen pipeline on 475 calls from late 2025. Stages, checks and confirmatory tests were recorded in a dated decision log, released with the protocol, before the corresponding results existed.

2 Related Work

Earnings calls and volatility. Qin and Yang (2019) introduced the task and the EarningsCall corpus, predicting volatility over 3 to 30 days from the transcript and the audio. HTML (Yang et al., 2020) added a sentence-level transformer with a multi-task loss, MAEC (Li et al., 2020) a larger corpus with sentence-aligned audio features, VolTAGE (Sawhney et al., 2020a) and Sawhney et al. (2020b) graph and multimodal fusion, DialogueGAT (Sang and Bao, 2022) a dialogue graph over the Q&A, and KeFVP (Niu et al., 2023) knowledge-enhanced features. ECC Analyzer (Cao et al., 2024) and Sound of Risk (Chen et al., 2025) are the strongest recent results and come without code. All of these are scored on the label and split files that VolTAGE and KeFVP redistribute, which §3 audits. Three newer corpora matter here. FinCall-Surprise (Shu et al., 2026a) was the first open corpus with raw audio and ships no identities. Earnings25 (Jiang et al., 2026) postdates every model's training data. MM-FinEval (Shu et al., 2026b), which appeared after this study was complete, is larger and also ships audio, but its release has no ticker or date fields and none of the checks below. The identity recovery used here applies to it.

Identity, memorisation and lookahead. Yu et al. (2025) showed on their own corpus that transcript representations mostly tell the model which company is speaking, and that the company's past volatility beats every transcript model. Language models recover the company even from transcripts with the names removed (Lopez-Lira et al., 2025). Lookahead bias in language models now has diagnostics and cutoff-bounded models of its own (Gao et al., 2025; Yan et al., 2026; Benhenda, 2026), but no earnings-call pipeline had been scored on calls after its training cutoff. The post-cutoff test in §5 does that.

Baselines, timing and bias. Persistence, GARCH and HAR-RV (Corsi, 2009) are hard to beat and often missing from published comparisons (Gu et al., 2020), so every table here is anchored on them. Yu et al. (2026) time earnings releases from 8-K filings to predict returns. We use the same filings to place day 0, which no volatility label set had done. Audio features encode speaker gender (Wang et al., 2024; Lin et al., 2024), so we report the gender confound.

Reproducibility. A systematic review found that only 14% of reported NLP scores were reproduced exactly (Belz et al., 2021), and in financial NLP discontinued data APIs and dead repository links make matters worse (Tatarinov et al., 2025). The reproduction study here records, per paper, what ran as released and what had to be repaired.

3 The Shared Test Files and the Protocol

What the field evaluates on. The EarningsCall (EC) benchmark of Qin and Yang (2019) and the two MAEC cohorts are scored on split and label files that VolTAGE and KeFVP redistribute. We audited those files (Table 1). The VolTAGE and KeFVP test files are byte for byte the same file, so the whole line of work shares one label set that was never recomputed from prices. Five of the six boundaries between training, validation and test have a zero-day gap (the sixth has three days) while the targets reach 30 days past each call. On the time-ordered splits the field reports, 80.4% (EC), 44.2% (MAEC-15) and 55.7% (MAEC-16) of test calls are from companies that also appear in training. Any time-ordered split allows that. The problem is that no paper reports the company-disjoint case, so the confound of Yu et al. (2025) has never been measured on these benchmarks. The original paper defines its horizons in calendar days, but most later work describes them as trading days (21 sessions rather than 30 at τ=30). Finally, the shipped MAEC labels correlate only 0.52–0.87 with calendar-day targets we recompute from prices under the same call-date anchor (0.25–0.66 at τ=3). We could not reproduce them under any single stated rule.

Corpora and company identity. FinCall-Surprise (Shu et al., 2026a) is the primary corpus, with 2,688 calls from 2019–2021 and MP3 audio, transcripts and slides. It ships four fields per record and no ticker, date, CIK or speaker names. We recover the company and the date from the transcript text and the slide metadata, check them against price data, and release the table with its 80 manual corrections. 2,496 calls resolve (92.9% overall, 95.3% of the earnings-type calls), and a 60-call manual audit passes 60/60. **MAEC** (Li et al., 2020) is the secondary corpus, with about 3,443 calls from 2015–2018, sentence-aligned audio features and no raw audio. **Earnings25** (Jiang et al., 2026) has 514 S&P 500 calls from July–December 2025, with webcast MP3 at 64 kbps for 281 calls and at most 24 kbps for 232. It postdates every model's training data, and 475 of its calls map to a ticker and join to prices.

Benchmark / split	calls	overlap	embargo	label defect
EC (Qin & Yang)	560	80.4%	0 / 0 d	137 zeros
MAEC-15 (KeFVP)	765	44.2%	0 / 0 d	binary cols
MAEC-16 (KeFVP)	1400	55.7%	0 / 3 d	binary cols
ours, FinCall temporal	2200	91.3%	47 / 47 d	n/a
ours, FinCall ticker-disjoint	2433	**0.0%**	n/a	n/a
ours, MAEC temporal	2264	79.4%	101 / 48 d	n/a
ours, MAEC ticker-disjoint	2688	**0.0%**	n/a	n/a

Table 1: **Audit of the shared test files** (regenerated by `ecvol audit substrate` from checksummed copies of the VolTAGE and KeFVP files, which are identical). Overlap is the share of test calls whose company is in training, and embargo is the gap train→val / val→test. Every time-ordered split allows overlap, ours included, but no paper in this line reports the disjoint case. Shipped MAEC labels correlate 0.52–0.87 with our recomputed calendar-day, call-date-anchored targets at $\tau \geq 7$.

Targets. Daily adjusted prices come from a free source and were checked against Tiingo on 50 tickers. Let r_t be the return on session t, counted from day 0, the last session before the call's information is public. The main target is $v_{\text{post}}(\tau) = \ln \sqrt{\frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^2}$ for $\tau \in \{3, 7, 15, 30\}$, and $v_{\text{pre}}(\tau)$ is the same quantity over the sessions ending at day 0. Under the **trading-day** definition $n = \tau$. Under the **calendar-day** definition of Qin and Yang (2019) the window holds every session within τ calendar days, so $n \approx 0.7\tau$, and at τ=3 a Thursday or Friday call has at most one session. We drop any window with fewer than two sessions and record the reason. That removes 42.5% of FinCall's calls at τ=3 calendar days, for which the released label sets still carry a value. The two definitions are not interchangeable. v_{post} correlates only 0.78 between them at τ=3, and on the temporal splits the persistence MSE is 44–63% higher there under

calendar days. A published MSE therefore means little unless its definition is stated (Appendix Table 10). Every result below uses trading days, and the definition is a column in every result table. We add two targets that are harder to game with company identity. The **change** $\Delta v = v_{\text{post}} - v_{\text{pre}}$ subtracts the company's own level, and the **HAR-residual** subtracts a HAR-RV forecast. After joining to prices, FinCall has 2,433 calls with a target (97.5%) and MAEC 2,688 (78.6%, the rest lacking usable price history).

When the information became public. Almost no transcript states a clock time, so we take the moment the information became public from EDGAR. It is the acceptance time of the earliest 8-K filing with Item 2.02 (the earnings release) within three days before to one day after the call. That covers 89.3% of FinCall, 68.6% of MAEC and 99.8% of Earnings25. The rest fall back, in recorded order, to Earnings25's call-start time, the before/after-market flag of Yu et al. (2025), and an after-hours assumption. A release before the open makes the call date itself day 1. Compared with the call-date anchor that every released label set uses, **day 0 moves for 53% of FinCall, 56% of MAEC and 72% of Earnings25 calls**, almost always one session earlier. Under the published anchor, the session in which the market reacts therefore sits in the *pre* window for most calls. On the measured anchor the persistence MSE is up to 11% higher on FinCall, 25% on MAEC and 27% on Earnings25, and Δv gains a positive mean (on FinCall $+0.52$ at τ=3 and $+0.10$ at τ=30). The call-date targets ship as a labelled variant. On Earnings25, which records call start times, the filing precedes the call for 98.9% of calls (median 100 minutes). The call can still be running during session 1, so these targets describe a release-aligned event window. A forward-looking variant starts the post window one session later, so that no target return begins before the call has ended (true for all 476 Earnings25 calls).

Splits and release. *Temporal* splits (about 70/10/20 in date order) leave a 30-trading-day gap between segments so that no target window crosses a boundary. *ticker-disjoint* splits keep every validation and test company out of training. Their overlap is 0.0%, against 91.3% (FinCall) and 79.4% (MAEC) on our own temporal splits. Because they are not in date order, a *date-ordered company-disjoint* split applies both rules at once, at the cost of a smaller test set (97 FinCall calls). The harness fails any feature that reads price data after the call's as-of time, and leakage assertions run in CI. The release contains all of the above, the derived-feature archives, the harness, and every result CSV with its run record (resolved config, git commit, seeds and file hash). Raw audio and transcripts are never redistributed.

4 Models, Protocol and Checks

Models, from simplest to richest. We organise our own models in stages. **Stage 0** holds the classical forecasts, namely persistence ($\hat{v} = v_{\text{pre}}$), the training-set mean of the target (a constant), EWMA (λ=0.94), HAR-RV (Corsi, 2009) and GARCH(1,1). **Stage 1** is a LightGBM model on past-volatility features, transcript length, number of Q&A turns and a **company fixed effect**, our identity-only baseline. **Stage 2** uses the transcript through BGE-M3 (BAAI, 2024) embeddings pooled per section (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates and surface statistics. **Stage 3** uses the audio, resampled to 16 kHz mono. Its features are eGeMAPSv02 functionals (Eyben et al., 2010) and WavLM-Large (Chen et al., 2021) and emotion2vec+ (Ma et al., 2023) embeddings over 30-second windows, averaged per call. **Stage 4** combines modalities by gated fusion or by stacking (a ridge over the text, audio and Stage-1 predictions, fit on the validation set). From Stage 2 on, every model is reported with and without a block of past-volatility features (v_{pre} at all horizons plus the HAR-RV inputs). The prediction heads are a ridge regression, with its penalty chosen on validation, and a small MLP run with five seeds. The MLPs are unstable across seeds (standard deviation of R^2_{OOS} up to 6.8), so the main grid uses one deterministic model per stage, chosen by a rule recorded in the decision log. MLP cells are in Appendix A.

Metrics and tests. We report MSE on log realised volatility and, as the headline metric, R^2_{OOS} relative to persistence (Gu et al., 2020), defined as $1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}_{\text{persist}})^2$. Because Δv has a positive mean on the measured anchor, persistence (which predicts Δv=0) is a weak reference there, so every Δv claim is also compared with the constant model. Diebold–Mariano tests (Diebold and Mariano, 1995) with the small-sample correction compare per-call squared errors against persistence and against the previous stage. The confirmatory comparisons (each stage against Stage 1

on Δv across horizons, Stage 4 against the best single modality, and the published split against the controlled split for each reproduced model) are Holm-corrected. The central question is whether call content improves on the same model without it, so each content model is also compared, on the same calls, with the past-volatility ridge it extends. We report the relative MSE reduction with a 95% bootstrap interval that resamples companies and read it against a 2% margin fixed before the comparison was run. An interval above the margin is an improvement, one below it shows no meaningful benefit, and anything else is inconclusive. All other analyses are exploratory.

Identity checks. We use four. A *company-only* model predicts each company's training mean and shows how far identity alone gets. Two *swaps* replace each test call's transcript or audio, either with another call by the same company or with a random call from the corpus. If the same-company swap leaves a model's score unchanged, the model was reading identity, and if the random swap also leaves it unchanged, the model was reading nothing. A linear *probe* measures how much identity the frozen call embedding carries by trying to recover the ticker from it. Finally, every table shows ticker-disjoint results next to temporal ones. The command `ecvol audit predictions` runs the prediction-side versions of the company-only and swap checks on any file with one row per (call, ticker, date, split, horizon, truth, prediction). It also reports split integrity, seen versus unseen companies, the between-company share of prediction variance and company-clustered significance. A prediction-side swap moves the whole prediction, price inputs included, so it measures company structure in the predictions. How a model responds to a different call with its other inputs held fixed needs the input-level swaps, and so the model itself.

Reproductions and post-cutoff tests. Of seventeen papers in this line, one releases code that runs as released (Same-Company-Same-Signal), seven release code that needs repairs or inputs that were never shipped, and nine release nothing usable (Table 16). We rerun the five with usable code (HTML, KeFVP, Same-Company-Same-Signal, Sawhney et al., 2020b, DialogueGAT) *on their own data*, using the authors' code wherever it can be made to start and a line-by-line port where it cannot. Every patch is in a released ledger. We first compare with the published number and then rerun the same model under a gapped split and a ticker-disjoint split of the same data. Earnings25 allows two more tests, namely the frozen Stage-3 audio models rerun by bitrate, and the whole FinCall-trained pipeline scored, with no retraining, on calls that postdate every model's training cutoff.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
S0 train mean	0.165	0.158	0.060	0.088
S0 HAR-RV	0.444	0.372	0.138	−0.285
S0′ past-vol ridge	0.412	0.314	0.094	0.113
S1 GBDT (ticker FE)	0.340	0.249	−0.126	−0.948
S2 text ridge + vol	0.333†	0.285	0.080	−0.077†
S3 audio ridge + vol	0.353†	0.268†	0.033†	0.050†
S4 stack fusion	0.406	0.347	0.066	0.100
Date-ordered disjoint split (n=97)				
S0 train mean	0.152	0.147	0.080	0.119
S0 HAR-RV	0.446	0.384	0.209	−0.120
S0′ past-vol ridge	0.409	0.315	0.159	0.158
S1 GBDT (ticker FE)	0.300	0.047	−0.324	−1.063
S2 text ridge + vol	0.160†	0.169†	0.039†	−0.056†
S3 audio ridge + vol	0.312†	0.147†	0.057†	0.098
S4 stack fusion	0.339†	0.108†	−0.180†	−0.028†

Table 2: **The ladder on FinCall-Surprise.** R^2_{OOS} vs. persistence on Δv (test). Persistence is 0 by construction, and the train-mean intercept is the featureless floor. The second block keeps test companies out of training and the split in date order. One pre-registered canonical model per stage (the Stage-3 ridge uses WavLM and eGeMAPS). Each content cell is tested against the past-volatility ridge S0′ on the same calls. A dagger marks a 95% interval of the relative MSE reduction, resampled by company, that lies below the pre-registered 2% margin, an asterisk one that lies above it, and an unmarked cell is inconclusive (Appendix Table 9). The full grid, MAEC, level-v, MSE and the MLP heads are in Appendix A.

5 Results

The tables in this section use FinCall-Surprise, the Δv target and the trading-day definition. MAEC, the level target, MSE, the calendar-day definition and every other model are in Appendix A and in the supplement, regenerated from the released CSVs.

The baseline to beat. HAR-RV is strong everywhere except one cell (Table 2). On the FinCall temporal split at τ=30, the training period includes the COVID spike and the test year (2021) is calm. There, every long-horizon forecast of the volatility level does worse than persistence, ours and every content model alike. The same models do fine on the ticker-disjoint split and on MAEC, so we treat this as a regime effect and note it. HAR-RV

Identity probe (linear, frozen call embedding)	accuracy	×chance
BGE-M3 text, FinCall (357 tickers)	89.5%	319×
BGE-M3 text, MAEC (902 tickers)	53.9%	487×
WavLM audio, FinCall (356 tickers)	76.3%	271×
emotion2vec+ audio, FinCall (356 tickers)	29.4%	105×

Content swap (R^2_{OOS} on Δv)	τ=3	τ=7	τ=15	τ=30
Text ridge, temporal split				
real	0.182	0.202	0.044	−0.133
same-company swap	0.174	0.194	0.047	−0.110
random swap	0.145	0.110	0.008	−0.152
Text ridge, date-ordered company-disjoint				
real	0.122	0.133	0.009	−0.092
same-company swap	0.136	0.139	0.019	−0.057
random swap	0.173	0.166	0.084	0.099
WavLM ridge + past-vol, ticker-disjoint split				
real	0.456	0.390	0.267	0.222
same-company swap	0.453	0.373	0.229	0.152
random swap	0.453	0.323	0.164	0.099

Table 3: **Identity checks.** The top block shows that frozen embeddings identify the company far above chance. The bottom block shows that where companies recur (temporal), giving a test call another call by the *same* company barely changes the text model, while a random call hurts. On the date-ordered company-disjoint split the real transcript scores no better than either swap. Audio swaps were run only on the ticker-disjoint split, which is not in date order.

is also not the strongest classical forecast, since GARCH(1,1) beats it at every temporal horizon (Appendix Table 6). The reference for every content claim below is the past-volatility ridge S0′, which sees the same price inputs as each content model and nothing else.

Text models mostly read the company (Table 3, Figure 2). Adding text features to the past-volatility ridge makes it *worse* in all eight cells of Table 2. The paired test finds no meaningful benefit in six and no improvement in any. A linear probe recovers the ticker from the frozen call embedding 89.5% of the time on FinCall and 53.9% on MAEC. On the temporal split, where test companies also appear in training, giving each test call the transcript of *another call by the same company* changes the score by less than 0.03, while a random transcript hurts. The model is reading which company is speaking. On the ticker-disjoint split the same-company swap does hurt at $\tau \geq 15$, which suggests some call-specific sensitivity, but that split is not in date order. On the date-ordered company-disjoint split the real transcript scores no better than either swap. On MAEC's temporal split the same-company swap also moves the score by 0.02 or less.

Audio adds nothing to past volatility. Audio-only ridges on eGeMAPS, WavLM-Large and emotion2vec+ stay within 0.10 of the constant at every horizon (Appendix Table 8) and never beat it by more than 0.02. Adding WavLM and eGeMAPS to the past-volatility ridge makes it worse in all eight cells, with no meaningful benefit in seven. In the audio-swap check (Table 3, bottom), the model with past volatility keeps its score at τ=3 when every test call is given a *random other company's* audio (0.456 to 0.453). At longer horizons the swap costs more. Even so, the model with real audio stays below the ridge with past volatility alone, so whatever the audio contributes does not add to the baseline. A probe still recovers the ticker from pooled WavLM embeddings 76.3% of the time. A gender check (dominant-speaker pitch proxy) finds no error gap (test MSE 0.243 against 0.239). Rerun on Earnings25 by bitrate, the gain from real over randomly swapped audio never exceeds 0.03, is negative in 7 of 12 cells, and is no larger at 64 kbps than at lower bitrates (Appendix Table 14). The strata differ in calls as well as bitrate, so this shows consistency across the groups observed, not a controlled codec test.

Combining modalities. Gated fusion is below the ridge heads with past volatility in 15 of 16 temporal and ticker-disjoint cells. The stacking model is the best content model. Against the past-volatility ridge it shows no meaningful benefit in all four date-ordered company-disjoint cells and is inconclusive in the four temporal ones. Its one apparent gain is on the ticker-disjoint split at τ=30, which is not in date order. There it reduces MSE by 11% against the past-volatility ridge, with an interval from 1% to 20% that is inconclusive under the margin, and by 8% against HAR-RV, with an interval that includes zero (Appendix Table 9). The forward-looking target lifts that cell above the margin, while on the date-ordered split the same model is 25% worse. The condition we set in advance for a positive paper, gains surviving the checks on at least two of four horizons, was not met.

Published models under the checks (Table 4, Figure 3). Among the five papers we could run, the results that reproduce belong to price-only or same-company models (Appendix A.4 separates direct reproductions, repaired runs and ports). Same-Company-Same-Signal reproduces from its shipped data (TSMixer to 10^{-5}, TMLP within 0.03), and its training-free same-company history comes within 0.05 of KeFVP's published MSE on EC and MAEC-15. KeFVP, a price-series model

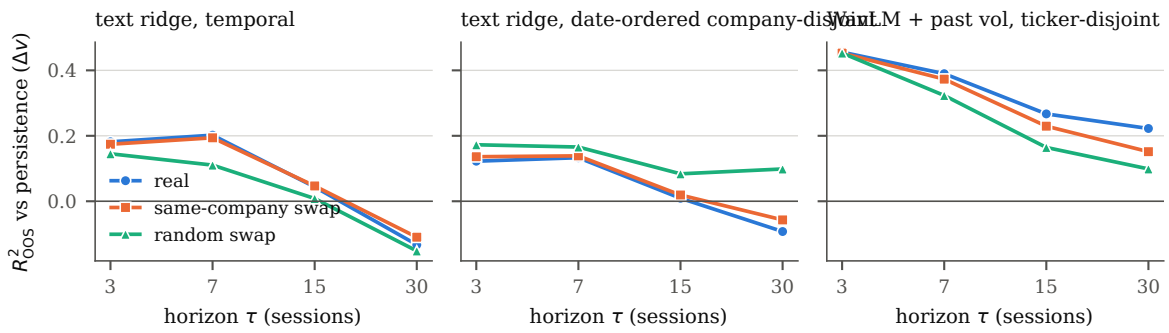


Figure 2: **Content swaps.** R^2_{OOS} vs. persistence on Δv (test) when each test call keeps its own transcript or audio (real), receives another call by the same company, or receives a random call. Where companies recur in training (left), the same-company swap leaves the text model unchanged. On the date-ordered company-disjoint split (middle) the real transcript does no better than either swap. The audio model with past volatility (right, ticker-disjoint split) keeps its score under a random swap at τ=3. Data from Table 3.

conditioned on text, reproduces seven of eight MAEC cells. HTML's released code does not reproduce its published table, and the published ensemble of Sawhney et al. (2020b) is matched only through its finance branch, with mixing weights chosen on the test set. DialogueGAT's corpus was never released, so it is ported to FinCall. Holding companies out raises the MSE of the three text models by 12–82%, while TSMixer and KeFVP move by −10% to +7% with no consistent sign. The SCSS and KeFVP comparisons use the same test calls. The HTML and ensemble ones do not, so they also reflect a change of test set. Training-set size does not explain the difference, since the held-out training sets are as large (EC) or cut by the same third (DEC, MAEC) for both groups, and the training-test gap moves models by at most 13%. HTML's own trained model changes by at most 4% when a test transcript is swapped for another call by the same company, against 9–23% for a different company's call.

Model (data), change in test MSE	τ=3	τ=7	τ=15	τ=30
Companies held out (ticker-disjoint vs. the model's own split)				
HTML text (EC)	+18%	+55%	+55%	+82%
HTML text+audio (EC)	+20%	+44%	+61%	+70%
Sawhney ensemble (EC)	+13%	+28%	+41%	+38%
SCSS TMLP, text (DEC)	+12%	+23%	+25%	+27%
SCSS TSMixer, price (DEC)	+7%	+5%	−6%	−8%
KeFVP, text+price (MAEC-15)	−3%	+4%	+3%	−1%
KeFVP, text+price (MAEC-16)	−9%	−3%	−10%	−4%
30-session embargo vs. the model's own split				
HTML text (EC)	+10%	+3%	−3%	+11%
Sawhney ensemble (EC)	+5%	−6%	−2%	−9%
SCSS TMLP (DEC)	−4%	+1%	0%	+3%
HTML text, transcript replaced by (own split, 97 calls)				
another call, *same* company	+1%	+4%	+3%	+2%
a call of a *different* company	+9%	+11%	+21%	+23%

Table 4: **The field's runnable models under our controls.** Each row keeps the configuration of Appendix Table 12 and changes only the split. Holding companies out costs every text model 12–82% while the price-dominated models show no consistent change, with held-out training sets as large (EC) or cut by the same third (DEC, MAEC). Absolute values are in Appendix Table 13.

After the cutoff (Table 5). We scored the FinCall-trained models, unchanged, on 475 Earnings25 calls from late 2025, after every model's training cutoff and on companies none had seen. On the level target, no model that was competitive before the cutoff got worse. HAR-RV rose from 0.44, 0.37, 0.14 and −0.29 to 0.60, 0.54, 0.35 and 0.08, and the content ridges rose with it (only the fusion ridge is unstable, at τ=30). Levels are not comparable across the cutoff, since the years and companies differ, but the ordering is, and it holds. HAR-RV beats every content model at every horizon, and no content model that was competitive before the cutoff collapses after it. Refit under the call-date anchor of the published label sets, the same models collapse before the cutoff. Text plus past volatility on Δv falls to between −1.7 and −3.6 at 3 to 15 days (Appendix Table 15), because the reaction session then sits inside the past-volatility features.

6 Discussion

Across every model we trained, the same picture holds. HAR-RV and the other classical forecasts are hard to beat, and scoring against persistence exposes a regime failure at the 30-day horizon that raw MSE would have hidden. Text features do

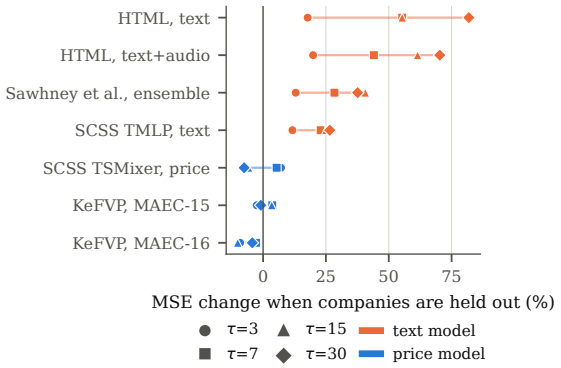


Figure 3: **Published models with companies held out.** Change in test MSE when the same model is scored on a ticker-disjoint split of its own data, one marker per horizon (data from Table 4). The MSE of the three text models rises by 12–82%, and the two price-based models show no consistent change.

Model	cutoff	τ=3	τ=7	τ=15	τ=30
S0 train mean	before	0.36	0.14	−0.37	−1.05
	after	0.58	0.47	0.15	−0.29
S0 HAR-RV	before	0.44	0.37	0.14	−0.29
	after	0.60	0.54	0.35	0.08
S2 text ridge + vol	before	0.43	0.34	0.03	−0.45
	after	0.57	0.50	0.25	−0.20
S3 WavLM ridge + vol	before	0.38	0.24	−0.09	−0.66
	after	0.57	0.43	0.15	−0.32
S4 fusion ridge + vol	before	0.41	0.32	−0.02	−0.63
	after	0.50	0.46	0.20	−1.93

Table 5: **Frozen-pipeline post-cutoff evaluation.** Heads fit once on FinCall's training split and scored unchanged before the cutoff (the 2021 FinCall test year) and after it (475 Earnings25 calls, late 2025), on the level target, R^2_{OOS} vs. persistence. The ordering survives the cutoff. Bootstrap confidence intervals and the Δv rows are in the released CSV.

not improve on past volatility alone, and the swap checks say why. A probe reads the ticker out of the call embedding nine times in ten, and where a company recurs in training, handing the model a different call by that company moves its score by less than 0.03. Audio behaves the same way at 40 kbps and at 64 kbps, identifying the company without improving the forecast, and the one stacking gain over the matched model appears only on a split that is not in date order. The shared test files could not have shown any of this, because they carry no company identity for the audio corpus, no company-disjoint split and no measured day 0. Applied to five published models on their own data, the checks give the same answer. The price-based models keep their accuracy when companies are held out, and the error of the text models rises by 12–82%. The two strongest recent positive results (Chen et al., 2025; Cao et al., 2024) could not be checked, since their code is not released, though the `audit predictions` command would make the company-level part of that check a matter of minutes for anyone holding a file of predictions. Transcripts and audio may well carry information about a company's risk. The test files in use cannot separate that information from knowing which company is speaking. Once they can, none of the content models evaluated here improves on a matched past-volatility model by the margin we fixed in advance. That is the bar the next model has to clear.

7 Recommendations and Outlook

The results above say what an evaluation has to control for before a content effect can be claimed, not that earnings calls carry no information. We suggest that future work on this task report six things. A **company-disjoint** split next to the temporal one. A **gap** between training and test dates of at least the longest horizon. **Past-volatility baselines** as the reference for every claim. The **same-company and random swaps**, or the released audit command. **Day 0 measured** from the release time. And the **horizon definition** stated with every number. Appendix Table 17 gives the reason for each and where the release provides it. ECVOL-bench covers all six for FinCall-Surprise and MAEC, the audit command applies the checks to any prediction file, and a quickstart script goes from the released archive of a corpus to the past-volatility baseline, a prediction file and the audit report in a few seconds, so a new model meets the same floor and checks on its first day.

There is also reason to expect that content signal can be found once it is looked for in the right place. Measured release times make **event-window targets** possible, where language about guidance and surprise should matter most and same-company history explains least. **Structured features** such as guidance changes can be audited one field at a time, unlike pooled embeddings, and combined with the fast numeric signal of Yu et al. (2026). **Audio** deserves a test on source-quality recordings with resolved speakers, and **larger corpora** such as MM-FinEval (Shu et al., 2026b) are open to the identity recovery used here. Each can be evaluated against the released baselines on the first day, which is what the protocol is for.

Limitations

One post-cutoff quarter. The post-cutoff test rests on 475 calls from a single calm quarter. Period, companies and source all change at once, so it is a temporal external validation of the ordering of models, not a measurement of pretraining contamination. **Small date-ordered test set.** The split that is both date-ordered and company-disjoint keeps 97 FinCall test calls, and the intervals resample companies but not calendar time, so shocks common to one quarter are not accounted for. **Forward-looking target.** It assumes a call after a pre-open release ends before that day's close, checked on Earnings25 only, and it was run on FinCall for the canonical models only. **Reproductions run on their own data**, not on ECVOL-bench, to keep "the model fails" separable from "the model does not transfer". Two carry known costs. KeFVP's released EC embeddings are unavailable upstream, and DialogueGAT's corpus was never released, so our port stops at its protocol. **MLP heads are seed-unstable** (standard deviation of R^2_{OOS} across seeds up to 6.8 on the level target and 1.1 on Δv), which is why the grid uses deterministic models. MLP cells in the appendix are seed means with that caveat. **Short-horizon R^2_{OOS} inflation.** The persistence denominator is noisiest at τ=3 (its MSE is about 2.8 times that at τ=7), so short-horizon gains are easier to show. **Timestamps.** 10.7% of FinCall and 31.4% of MAEC calls (13.0% of the MAEC calls that carry targets) have no Item 2.02 filing in the window and use a recorded fallback tier, and intraday releases are treated as after-hours. Results under the call-date anchor of prior label sets are not interchangeable with ours, and the call-date targets are shipped as a variant. **Regime sensitivity.** FinCall-Surprise spans COVID. The temporal τ=30 failure shows that long-horizon level forecasts are regime-sensitive, which is why we emphasise Δv and per-year breakdowns. **MAEC coverage.** Its 78.6% price-join rate (2015–2018 delistings) makes it a secondary corpus, and it ships no raw audio. **Source audio.** Every FinCall file is a 40 kbps MP3 transcode and Earnings25 is 64 kbps or lower, so no result here is established on lossless audio. **No diarisation.** Audio is pooled per call, and the gender analysis uses a dominant-speaker pitch proxy. **Identity reconstruction is imperfect.** 7.1% of FinCall calls remain unresolved and are excluded with a reason code. **No fine-tuned or LLM-based content model of our own.** Stages 2 to 4 use frozen encoders with ridge and MLP heads. The fine-tuned HTML reproduction shows the same dependence on company identity (Table 4), and §7 points to structured features extracted by language models as the place to look, but this paper does not test such a model. The audit command lets anyone check one from its prediction file.

Ethical Considerations

Data and licensing. We use openly licensed corpora (FinCall-Surprise under Apache-2.0, MAEC under CC-BY-SA-4.0, Earnings25 under CC-BY-4.0) and open-weight models. Derived features inherit and state their upstream licence, and raw audio or transcripts from sources with unclear terms (EC, DEC, the reproduced repositories) are never redistributed. Calls are public corporate disclosures, and we release no speaker-level audio or identity, only pooled per-call features. **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022), so we report a gender-confound diagnostic. **Intended use.** This is a research tool, not investment advice. Predictions are noisy and regime-sensitive and should not drive automated trading. **Reproducibility.** Given the documented reproducibility problems in financial NLP (Tatarinov et al., 2025), the environment is pinned, every data file is checksummed, every result CSV regenerates byte for byte from its command and run manifest, and the curated tables copy cells from those CSVs.

References

BAAI. 2024. BGE-M3 sentence embeddings. Hugging Face (MIT).

Anya Belz, Shubham Agarwal, Anastasia Shimorina, and Ehud Reiter. 2021. A systematic review of reproducibility research in natural language processing. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume (EACL)*, pages 381–393.

Mostapha Benhenda. 2026. Look-ahead-bench: a standardized benchmark of look-ahead bias in point-in-time LLMs for finance. arXiv:2601.13770.

Yupeng Cao et al. 2024. ECC analyzer: Extracting trading signal from earnings conference calls using large language models for stock volatility prediction. In *Proceedings of the 5th ACM International Conference on AI in Finance (ICAIF)*.

Sanyuan Chen et al. 2021. WavLM: Large-scale self-supervised pre-training for full stack speech processing. Hugging Face (MIT): microsoft/wavlm-large.

Xiaoliang Chen et al. 2025. The sound of risk: A multimodal physics-informed acoustic model for forecasting market volatility and enhancing market interpretability. arXiv:2508.18653.

Fulvio Corsi. 2009. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*.

Francis X. Diebold and Roberto S. Mariano. 1995. Comparing predictive accuracy. *Journal of Business & Economic Statistics*.

Florian Eyben, Martin Wöllmer, and Björn Schuller. 2010. openSMILE / eGeMAPS acoustic feature extraction. github.com/audeering/opensmile-python.

Zhenyu Gao, Wenxi Jiang, and Yutong Yan. 2025. Detecting lookahead bias in LLM forecasts. arXiv:2512.23847.

Shihao Gu, Bryan Kelly, and Dacheng Xiu. 2020. Empirical asset pricing via machine learning. *The Review of Financial Studies*.

Denglin Jiang, Haoran Zhou, Anshul Wadhawan, Brendan Fahy, Vinay Ramesh, David Weisberg, Dmitriy Derkachevskiy, Helen Sheehan, Srivas Prasad, and Michele Franceschini. 2026. Earnings25: A comprehensive 500-hour speech benchmark for finance. Interspeech 2026; arXiv:2607.23813; Zenodo record 10.5281/zenodo.18762168 (CC-BY-4.0).

Jiazheng Li, Linyi Yang, Barry Smyth, and Ruihai Dong. 2020. MAEC: A multimodal aligned earnings conference call dataset for financial risk prediction. In *Proceedings of the 29th ACM International Conference on Information and Knowledge Management (CIKM)*.

Yi-Cheng Lin, Haibin Wu, Huang-Cheng Chou, Chi-Chun Lee, and Hung-yi Lee. 2024. Emo-bias: A large scale evaluation of social bias on speech emotion recognition. In *Proceedings of Interspeech 2024*, pages 4633–4637. ArXiv:2406.05065.

Alejandro Lopez-Lira, Yuehua Tang, and Mingyin Zhu. 2025. The memorization problem: Can we trust LLMs' economic forecasts? arXiv:2504.14765.

Ziyang Ma et al. 2023. emotion2vec: Self-supervised pre-training for speech emotion representation. Hugging Face: emotion2vec/emotion2vec_plus_large.

Puneet Mathur et al. 2022. DocFin: Multimodal financial prediction and bias mitigation using semi-structured documents. In *Findings of the Association for Computational Linguistics: EMNLP 2022*.

Hao Niu et al. 2023. KeFVP: Knowledge-enhanced financial volatility prediction. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

ProsusAI. 2019. FinBERT: Financial sentiment analysis with bert. Hugging Face.

Yu Qin and Yi Yang. 2019. What you say and how you say it matters: Predicting stock volatility using verbal and vocal cues. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*.

Yunxin Sang and Yang Bao. 2022. DialogueGAT: A graph attention network for financial risk prediction by modeling the dialogues in earnings conference calls. In *Findings of the Association for Computational Linguistics: EMNLP 2022*.

Ramit Sawhney, Piyush Khanna, Arshiya Aggarwal, Taru Jain, Puneet Mathur, and Rajiv Ratn Shah. 2020a. VolTAGE: Volatility forecasting via text audio fusion with graph convolution networks for earnings calls. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.

Ramit Sawhney, Puneet Mathur, Ayush Mangal, Piyush Khanna, Rajiv Ratn Shah, and Roger Zimmermann. 2020b. Multimodal multi-task financial risk forecasting. In *Proceedings of the 28th ACM International Conference on Multimedia (MM)*.

Dong Shu, Yanguang Liu, Huopu Zhang, and Mengnan Du. 2026a. FinCall-Surprise: A large scale multi-modal benchmark for earning surprise prediction. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13357–13370. Data: github.com/Tizzzzy/FinCall-Surprise (Apache-2.0).

Dong Shu, Yanguang Liu, Huopu Zhang, Saisai Hu, Haiyan Zhao, Hekun Huang, and Mengnan Du. 2026b. MM-FinEval: A multi-task multimodal benchmark for real-world financial forecasting. arXiv:2609.38523.

Nikita Tatarinov et al. 2025. Language modeling for the future of finance: A survey into metrics, tasks, and data opportunities. arXiv:2504.07274.

Wang et al. 2024. AMA-LSTM: Pioneering robust and fair financial audio analysis for stock volatility prediction. NAACL 2024 Industry Track; arXiv:2407.18324.

Yutong Yan et al. 2026. DatedGPT: Preventing lookahead bias in large language models with time-aware pretraining. arXiv:2603.11838.

Linyi Yang, Tin Lok James Ng, Barry Smyth, and Ruihai Dong. 2020. HTML: Hierarchical transformer-based multi-task learning for volatility prediction. In *Proceedings of The Web Conference (WWW)*.

Ding Yu, Zhuo Liu, and Hangfeng He. 2025. Same company, same signal: The role of identity in earnings call transcripts. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 18403–18422.

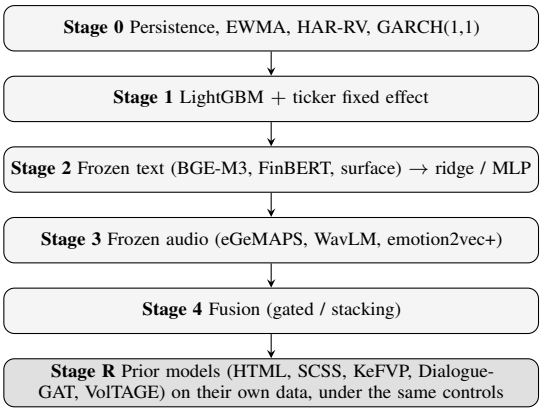


Figure 4: **The modelling ladder.** From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Every stage runs through the identity checks. The darker box is the reproduction track, which applies the same checks to published models on their own data.

Ding Yu, Zhuo Liu, Hao Zhang, and Hangfeng He. 2026. Fast numbers, slow language: Bridging quantitative and qualitative earnings signals. arXiv:2606.29734.

A Extended Results

This appendix holds the full set of models behind each main-text table, the horizon-definition comparison, the reproduction study in detail, the Earnings25 bitrate study, the call-date-anchor variant of the post-cutoff test, and the code-availability audit. The complete generated grids (Result Tables 1–4, every corpus, target, metric, split and horizon) are in the separate supplement. Every number traces to a released result CSV with a run record, and the curated tables copy cells from those CSVs (`REPRODUCE.md` lists the command behind each).

A.1 All models of Stages 0–3

Tables 6, 7 and 8 give the full Stage-0/1, Stage-2 and Stage-3 model sets on the Δv target, including the past-volatility-only and MLP heads omitted from Table 2.

Figure 5 plots the canonical model of each stage from Table 2 against the train-mean constant.

A.2 Increment over the matched price-only model

Table 9 gives the pre-registered paired comparison behind the markers of Table 2, on all three splits and under both the measured target and the forward-looking one. The forward-looking target removes most of the positive drift of Δv (its mean at τ=3 falls from 0.51 to 0.12). Its only improvements over the matched model are the stack at τ=30 on the ticker-disjoint split, which is not in date order, and none of the content models improves on it on the date-ordered company-disjoint split.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*
Ticker-disjoint split (n=484)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table 6: **Baselines.** R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal τ=30 cell. Asterisks mark Diebold–Mariano significance vs. persistence ($p < 0.05$).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (text)	0.182*	0.202	0.044*	-0.133*
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ticker-disjoint split (n=484)				
Ridge (text)	0.195*	0.217*	0.195*	0.169
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*

Table 7: **Text content (Stage 2).** R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head, and adding text to past volatility does not help. Asterisks mark Diebold–Mariano significance vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

A.3 Horizon definition

Table 10 compares the trading-day and calendar-day definitions of the horizon on FinCall and MAEC, both at the level of the targets (sessions per window, correlation between the two definitions) and at the level of the Stage-0 results (persistence MSE and HAR-RV R^2_{OOS} under each definition).

A.4 Reproduction study

Table 11 is the first entry of the study, the HTML head over our frozen BGE-M3 chunk sequences on EC under the published split, a gapped split

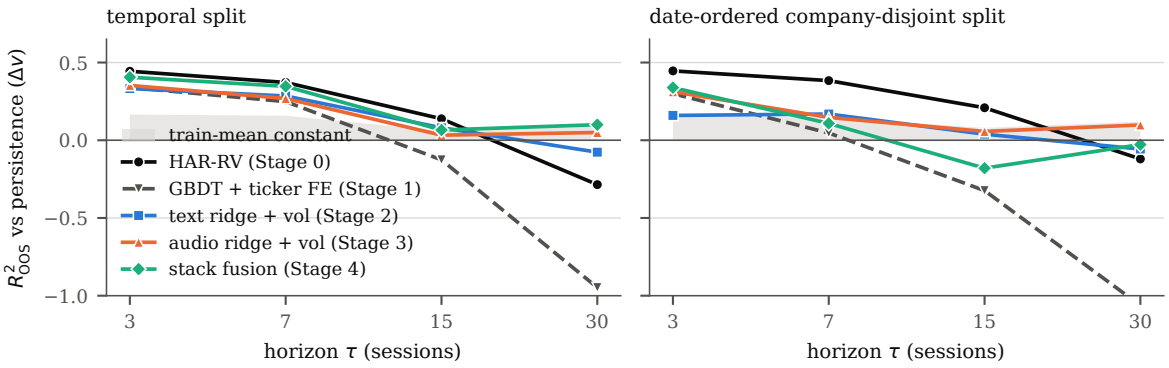


Figure 5: **The ladder.** R^2_{OOS} vs. persistence on Δv (FinCall test) for the canonical model of each stage. The shaded band is the train-mean constant. Content stages clear HAR-RV only at τ=30, where HAR-RV itself falls below persistence on both splits, and none clears the past-volatility ridge there. Data from Table 2.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (eGeMAPS)	0.163*	0.156*	0.059*	0.079*
Ridge (WavLM)	0.156*	0.148*	-0.027	-0.008*
Ridge (emotion2vec+)	0.155*	0.129*	0.022*	0.027*
Ridge (WavLM+past-vol)	0.357	0.262	-0.044	0.052*
Ticker-disjoint split (n=484)				
Ridge (eGeMAPS)	0.203*	0.119*	0.137*	0.044*
Ridge (WavLM)	0.178*	0.192*	0.131*	0.026*
Ridge (emotion2vec+)	0.181*	0.185*	0.122*	0.047*
Ridge (WavLM+past-vol)	0.456	0.390	0.267	0.222
Audio-shuffle control of WavLM+past-vol (ticker-disjoint)				
Real audio	0.456	0.390	0.267	0.222
Within-ticker shuffle	0.453	0.373	0.229	0.152
Global shuffle	0.453	0.323	0.164	0.099

Table 8: **Audio content (Stage 3).** R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads stay within 0.10 of the train-mean constant and never beat it by more than 0.02. The head with past volatility is below the past-volatility-only ridge of Table 7 in every cell, and at τ=3 it keeps its score when every test call receives a random other company's audio (bottom block), so past volatility rather than audio content carries it. Asterisks mark Diebold–Mariano significance vs. the Stage-1 ticker-FE GBDT ($p < 0.05$, read the direction from the value). Full head set in Appendix A.

and a ticker-disjoint split with identical labels. Table 12 reruns the five papers with usable code on their own data with the authors' code wherever it runs and a line-by-line port where it cannot. The released fidelity ledger records every patch and substitution. Table 13 gives the absolute values behind Table 4, and Table 16 the code-availability audit of the seventeen papers.

Same-Company-Same-Signal reproduces. Its TSMixer baseline matches the 76 published cells to a median 10^{-5} from the shipped data alone. Its TMLP text model on the authors' OpenAI embeddings matches per-window means within 0.03 and recovers the paper's finding that embeddings swapped within a company do as well as the real ones. Its training-free baselines on EC and MAEC reproduce all 48 published cells at the notebook's rounding. **HTML** does not. The released driver splits features and labels with three independent unseeded shuffles. With the authors' model classes unchanged on rebuilt inputs, the published recipe (ten epochs, learning rate 2×10^{-5}, raw log-volatility targets) beats the published number only at τ=3 and τ=7 on the shipped code's random split with dropout 0, reaches the published 15- and 30-day values on no split, and on the VolTAGE split every later paper inherited it is worse than persistence at every horizon except τ=3. Its text+audio row reproduces under the shipped code's random split and under no other. **Sawhney et al.** cannot run as released (an unshipped call inventory, price files and lexicons, and a TensorFlow 2.1 stack). Ported, the finance SVR alone gives 0.712/0.370/0.218/0.154, the text and audio branches are untrained after the two and one epochs the scripts prescribe, and the published ensemble is reproduced only when the mixing weights are chosen on the test set, as the released script does, where an untrained branch's near-constant output acts as an intercept on the finance branch. **DialogueGAT**'s corpus and labels were never released. Ported onto FinCall's speaker turns under the paper's per-year protocol, it beats persistence at short horizons and collapses on the 2020 split. **KeFVP**'s released script needed five release-defect patches before it would start. Its MAEC rows use sentence embeddings regen-

Target, split	Model	τ=3	τ=7	τ=15	τ=30
Measured anchor, temporal	HAR-RV	+5 [+3, +8]*	+8 [+6, +11]*	+5 [+2, +8]*	−45 [−61, −31]†
	S2 text ridge + vol	−14 [−25, −3]†	−4 [−15, +6]	−2 [−12, +8]	−21 [−32, −12]†
	S3 audio ridge + vol	−10 [−18, −3]†	−7 [−15, +0]†	−7 [−14, −1]†	−7 [−12, −2]†
	S4 stack fusion	−1 [−7, +4]	+5 [−0, +10]	−3 [−9, +2]	−1 [−6, +4]
Measured anchor, date-ordered disjoint	HAR-RV	+6 [+0, +11]	+10 [+3, +16]*	+6 [−6, +17]	−33 [−64, −7]†
	S2 text ridge + vol	−42 [−90, −12]†	−21 [−45, −4]†	−14 [−34, +1]†	−25 [−49, −5]†
	S3 audio ridge + vol	−16 [−38, −2]†	−25 [−42, −10]†	−12 [−24, −1]†	−7 [−21, +6]
	S4 stack fusion	−12 [−32, +0]†	−30 [−50, −16]†	−40 [−64, −21]†	−22 [−45, −3]†
Measured anchor, ticker-disjoint	HAR-RV	+4 [+0, +8]	+4 [+1, +8]	+1 [−3, +4]	+4 [−0, +7]
	S2 text ridge + vol	−34 [−53, −20]†	−18 [−29, −6]†	−14 [−26, −4]†	−2 [−16, +8]
	S3 audio ridge + vol	−25 [−45, −9]†	−20 [−37, −6]†	−16 [−33, −3]†	−15 [−31, +1]†
	S4 stack fusion	−5 [−13, +2]	−1 [−11, +8]	−1 [−13, +9]	+11 [+1, +20]
Forward-looking, temporal	HAR-RV	+12 [+9, +14]*	+14 [+11, +17]*	+7 [+3, +10]*	−67 [−87, −49]†
	S2 text ridge + vol	−32 [−47, −19]†	−23 [−35, −11]†	−10 [−21, −0]†	−35 [−48, −23]†
	S3 audio ridge + vol	−24 [−47, −8]†	−20 [−32, −10]†	−10 [−19, −2]†	−5 [−11, +1]†
	S4 stack fusion	−3 [−9, +3]	−5 [−12, +1]†	−18 [−27, −10]†	−1 [−6, +4]
Forward-looking, date-ordered disjoint	HAR-RV	+9 [+2, +16]	+12 [+5, +19]*	+14 [+4, +22]*	−34 [−68, −6]†
	S2 text ridge + vol	−63 [−112, −30]†	−34 [−60, −12]†	−24 [−39, −12]†	−32 [−54, −12]†
	S3 audio ridge + vol	−32 [−63, −12]†	−33 [−58, −13]†	−16 [−28, −5]†	−5 [−20, +7]
	S4 stack fusion	−13 [−35, +4]	−75 [−115, −43]†	−97 [−134, −64]†	−25 [−46, −7]†
Forward-looking, ticker-disjoint	HAR-RV	+4 [−1, +8]	+3 [−2, +8]	−1 [−6, +3]	+4 [−0, +7]
	S2 text ridge + vol	−47 [−65, −31]†	−18 [−32, −7]†	+0 [−13, +11]	+7 [−3, +16]
	S3 audio ridge + vol	−25 [−39, −13]†	−16 [−31, −5]†	−7 [−15, −0]†	−13 [−28, +0]†
	S4 stack fusion	−4 [−13, +3]	−2 [−12, +6]	+9 [+1, +15]	+16 [+8, +23]*

Table 9: **Increment over the matched price-only model.** Relative reduction of test MSE on Δv (%) against the past-volatility ridge on the same calls, with a 95% percentile bootstrap interval that resamples companies (1000 draws). Positive values favour the content model. The 2% margin and the reading rule were fixed before the table was computed. A dagger marks an upper bound below the margin (no meaningful benefit), an asterisk a lower bound above it (improvement), and unmarked cells are inconclusive. HAR-RV is shown against the same reference. The forward-looking target starts the post window one session later, so no target return begins before the call has ended.

erated with the authors' own generator because the released ones were never published, and seven of eight cells fall within one published standard deviation (the eighth, MAEC-16 at τ=15, is the row of their table that does not average to its own stated mean). On EC the released KePt embeddings are unavailable upstream (the linked folder is empty). With BERT-large pooler embeddings from the same generator, which is KeFVP without its knowledge-enhanced pre-training, the mean test MSE is 0.651/0.302/0.194/0.127 against the published 0.610/0.291/0.183/0.114, above all four published means and within one of our repeat standard deviations at only two horizons, so the row is not counted as reproduced.

Under the checks, the 30-session gap removes half to two thirds of KeFVP's MAEC training calls, so those rows cannot separate leakage from data loss and are left to the ledger. DialogueGAT's port (one seed, 42–60 held-out calls per year) shows no consistent direction under either check.

A.5 Audio by bitrate on Earnings25

Table 14 reruns the Stage-3 audio models on Earnings25, pooled and per bitrate stratum, with the random swap as the check.

A.6 Post-cutoff evaluation under the call-date anchor

Table 15 repeats the post-cutoff evaluation with day 0 anchored on the call date on both sides of the cutoff, the anchor every released label set uses, with the heads refit under that anchor. HAR-RV barely moves, but the ridge models that carry a learned past-volatility block collapse before the cutoff under that anchor, because the block absorbs the reaction session that the after-hours rule leaves inside the pre-window for before-open calls. On Δv at τ=30 the frozen WavLM+past-volatility model is the only content model whose post-cutoff confidence interval excludes zero under the measured anchor (0.18, CI 0.14–0.24, against HAR's 0.08, and the constant also scores 0.18 there). That is a single-quarter observation without a Diebold–Mariano test and we report it as exploratory.

A.7 Code availability

Table 16 is the code-availability audit of the seventeen papers in this line, regenerated by `ecvol audit code`.

	τ=3	τ=7	τ=15	τ=30
Targets (FinCall, calls ok under both conventions)				
Sessions in calendar window (mean)	2.4	4.9	10.8	21.2
Calls with $\geq$2 sessions (of 2,433)	1,398	2,433	2,433	2,433
corr($v_{\text{post}}^{\text{trad}}, v_{\text{post}}^{\text{cal}}$)	0.784	0.942	0.963	0.940
corr($\Delta v^{\text{trad}}, \Delta v^{\text{cal}}$)	0.751	0.887	0.904	0.891
Persistence MSE, level-v, temporal test (trading / calendar)				
FinCall	1.270 / 2.071	0.448 / 0.612	0.211 / 0.298	0.115 / 0.152
MAEC	1.774 / 2.553	0.712 / 0.986	0.356 / 0.492	0.201 / 0.263
HAR-RV R^2_{OOS} vs. persistence, level-v, temporal test (trading / calendar)				
FinCall	0.444 / 0.419	0.372 / 0.402	0.138 / 0.275	$-$0.285 / $-$0.053
MAEC	0.603 / 0.562	0.486 / 0.502	0.321 / 0.400	0.221 / 0.265

Table 10: Horizon definition. The calendar-day windows of Qin and Yang (2019) hold about 0.7τ sessions, and at τ=3 43% of the calls have at most one post-call session and are excluded. The persistence MSE, the leaderboard's metric, rises by 44–63% at τ=3 under calendar days, so published MSE values are not comparable across definitions. The FinCall temporal τ=30 deficit of HAR-RV against persistence (the COVID-regime cell, §5) holds under both definitions (-0.285 and -0.053). The full model and horizon grid under both definitions is in the supplement.

Condition (EC)	model	τ=3	τ=7	τ=15	τ=30
paper, published split, published labels	HTML (text, WWM-BERT)	1.175	0.372	0.153	0.133
published split, published labels	HTML head (BGE-M3)	0.769	0.436	0.285	0.223
	persistence	1.485	0.570	0.340	0.202
published split, our labels	HTML head (BGE-M3)	0.760	0.420	0.285	0.214
	persistence	1.826	0.590	0.333	0.183
embargoed temporal, our labels	HTML head (BGE-M3)	0.778	0.448	0.278	0.265
	persistence	1.948	0.629	0.353	0.192
ticker-disjoint, our labels	HTML head (BGE-M3)	**0.773**	**0.503**	**0.394**	**0.291**
	persistence	1.670	0.728	0.418	0.213

Table 11: **Reproduced prior model under leakage-proof evaluation.** The HTML head reproduces the paper's ordering against price baselines at τ=3 and beats the published number there, but not the published τ=15 and τ=30 values, which lie below the persistence floor of the shipped labels themselves. At τ=30 every condition is worse than persistence. Moving the same head and labels from the published split to a ticker-disjoint split raises test MSE by 2%, 20%, 38% and 36%. The embargoed temporal split costs 2%, 7%, $-$2% and 24%.

A.8 The reporting checklist

Table 17 expands the six items of §7 with the evidence behind each and the part of the release that provides it.

Paper / setting	what ran	ours τ=3	ours 7	ours 15	ours 30	published τ=3	published 7	published 15	published 30
HTML (Yang et al., 2020), EC, authors' model classes verbatim, inputs rebuilt (WWM-BERT layer −2), driver corrected									
as coded, random 70/10/20, dropout 0	text	0.660	0.336	0.253	0.160	1.175	0.372	0.153	0.133
as published, chronological 70/10/20, dropout 0.5	text	0.972	0.439	0.406	0.248				
published split (VolTAGE), dropout 0.5	text	1.231	1.122	1.063	1.479				
persistence on the shipped labels	n/a	1.485	0.570	0.340	0.202				
as coded, random 70/10/20, dropout 0	text + 27 Praat	0.658	0.347	0.244	0.165	0.845	0.349	0.251	0.158
as published, chronological 70/10/20, dropout 0.5	text + 27 Praat	0.876	0.520	0.291	0.239				
published split (VolTAGE), dropout 0.5	text + 27 Praat	2.701	1.195	1.436	1.411				
Same-Company-Same-Signal (Yu et al., 2025), DEC, authors' `run.py`*, per-window means over 19 rolling quarters*									
TSMixer (price history only)	76 cells	0.561	0.252	0.210	0.257	0.561	0.252	0.211	0.256
TMLP on OpenAI embeddings	76 cells	0.608	0.285	0.215	0.202	0.587	0.283	0.211	0.205
TMLP, embeddings shuffled within ticker	76 cells	0.572	0.276	0.213	0.209	0.565	0.273	0.204	0.200
TMLP, embeddings shuffled across all	76 cells	0.646	0.315	0.259	0.249	0.637	0.319	0.255	0.247
Same-Company-Same-Signal, EC / MAEC, authors' notebook, training-free, augmented same-ticker earnings history									
EC, Aug_STPEV	no training	0.569	0.293	0.201	0.122	0.569	0.293	0.201	0.122
MAEC-15, Aug_STPEV	no training	0.443	0.214	0.130	0.112	0.443	0.214	0.130	0.112
MAEC-16, Aug_STPEV	no training	0.470	0.238	0.168	0.113	0.470	0.238	0.168	0.113
KeFVP (Niu et al., 2023), authors' `final_series_infer.py` *(five release-defect patches), 10 repeats*									
MAEC-15, regenerated BERT-base embeddings	text + price	0.419	0.185	0.123	0.086	0.418	0.187	0.122	0.087
MAEC-16, regenerated BERT-base embeddings	text + price	0.442	0.288	0.363	0.188	0.445	0.279	0.303	0.177
EC, regenerated BERT-large embeddings (no KePt)	text + price	0.651	0.302	0.194	0.127	0.610	0.291	0.183	0.114
Sawhney et al. (2020b), EC, PyTorch port of the released scripts (their labels, split, branches); 5 seeds									
finance SVR (30 past volatilities)	port	0.712	0.370	0.218	0.154				
text BiLSTM, their 2 epochs	port	2.135	1.580	1.384	1.401				
aligned audio, their 1 epoch	port	6.514	5.295	5.368	5.130				
ensemble, (α, β) tuned on test (theirs)	port	0.639	0.298	0.175	0.102	0.601	0.308	0.181	0.119
ensemble, (α, β) tuned on validation	port	0.686	0.324	0.187	0.117				
persistence on their labels	n/a	1.491	0.563	0.345	0.205				
DialogueGAT (Sang and Bao, 2022), PyG port on FinCall turns (role-typed speakers); published cells are on their unreleased corpus									
FinCall 2019 (their 2015 protocol)	port	0.694	0.320	0.206	0.134	0.453	0.324	0.190	n/a
persistence	n/a	1.429	0.566	0.289	0.139				
FinCall 2020 (their 2016 protocol)	port	0.621	0.350	0.392	0.270	0.455	0.288	0.181	n/a
persistence	n/a	1.409	0.335	0.184	0.101				
FinCall 2021 (their 2017–18 protocol)	port	0.794	0.344	0.182	0.082	0.409	0.289	0.204	n/a
persistence	n/a	1.301	0.465	0.225	0.118				

Table 12: **Reproductions with the authors' own code and data.** Every cell regenerates from one `ecvol reproduce` command, and the fidelity ledger lists what is verbatim, patched, substituted or reason-coded per paper. Three of the five reproduce. TSMixer matches to 10^{-5}, TMLP within single-seed noise (including the paper's finding that real and within-company shuffled embeddings score alike), and KeFVP on MAEC within one published standard deviation on seven of eight cells with regenerated sentence embeddings. On EC, whose released KePt embeddings are unavailable upstream, the regenerated BERT-large row is above all four published means and within one of our repeat standard deviations at only two horizons, and is not counted. HTML's text-only recipe does not reach its published long-horizon cells on any split, while its text+audio row reproduces at $\tau \geq 7$ under the shipped-code condition (random split, dropout 0) and on no other. Sawhney's published ensemble is reachable only when the mixing weights are tuned on the test set, as the released code does, with untrained text and audio branches acting as an intercept on the finance SVR.

Model / condition	n_{tr}	n_{te}	τ=3	7	15	30
HTML, authors' classes (EC; seeds 0–2)						
text, shipped-code random split (anchor)	≈390	112	0.660	0.336	0.253	0.160
text, embargoed	285	101	0.726	0.347	0.246	0.178
text, ticker-disjoint	386	109	0.777	0.522	0.393	0.291
persistence on the ticker-disjoint test set	n/a	109	1.452	0.550	0.327	0.188
text+audio, anchor	≈390	112	0.658	0.347	0.244	0.165
text+audio, embargoed	285	101	0.684	0.345	0.232	0.186
text+audio, ticker-disjoint	386	109	0.789	0.500	0.394	0.281
transcript shuffle, anchor split, the 97 test calls whose company has another call						
text, real transcript	≈390	97	0.683	0.348	0.262	0.165
text, another call of the *same* company	≈390	97	0.693	0.361	0.269	0.169
text, a call of a *different* company	≈390	97	0.745	0.386	0.318	0.203
text+audio, real / same company / different company, τ=3	≈390	97	0.675 / 0.692 / 0.784			
Sawhney et al., port, ensemble with (α, β) tuned on validation (EC; 5 seeds)						
their split (anchor)	328	110	0.686	0.324	0.187	0.117
embargoed	283	100	0.719	0.306	0.184	0.107
ticker-disjoint	383	109	0.775	0.416	0.263	0.161
its own finance SVR, ticker-disjoint	383	109	0.700	0.365	0.256	0.158
Same-Company-Same-Signal, authors' `run.py` *(DEC; seed 2021; means over 19 rolling quarters)*						
TMLP (text), paired anchor	900	30	0.591	0.288	0.224	0.211
TMLP (text), ticker-disjoint	600	30	0.660	0.354	0.280	0.267
TMLP (text), embargoed (vs anchor 0.608 / 0.285 / 0.215 / 0.202)	895	90	0.585	0.287	0.216	0.208
TSMixer (price), paired anchor	900	30	0.519	0.241	0.206	0.252
TSMixer (price), ticker-disjoint	600	30	0.556	0.254	0.194	0.233
persistence on the held-out calls	n/a	30	1.258	0.457	0.252	0.224
KeFVP, authors' script (MAEC, regenerated BERT-base embeddings; 3 repeats)						
MAEC-15, paired anchor	535	48	0.470	0.192	0.145	0.114
MAEC-15, ticker-disjoint	353	48	0.458	0.199	0.150	0.113
MAEC-16, paired anchor	980	92	0.416	0.248	0.347	0.186
MAEC-16, ticker-disjoint	661	92	0.378	0.241	0.312	0.178
persistence on the MAEC-16 held-out calls	n/a	92	0.622	0.265	0.170	0.116

Table 13: **The reproduced models under our controls.** Each block keeps the configuration of Table 12 and changes only the split. Paired anchors score the unchanged model on exactly the ticker-disjoint test calls. Removing same-company history costs the three text models 12–82% (HTML, Sawhney, TMLP, where TMLP is worse in 16–18 of 19 quarters), while both price-dominated models (TSMixer, KeFVP) show no consistent change, although the ticker-disjoint training sets are as large (EC) or cut by the same third (DEC, MAEC) in both groups. The 30-session embargo moves HTML, Sawhney and SCSS by at most 13% in either direction. KeFVP's embargoed MAEC re-splits keep only a half or a third of its training calls, cannot separate leakage from data loss, and are reported in the ledger rather than here. DialogueGAT (one seed, 42–60 held-out calls per year) shows no consistent direction and is also left to the ledger.

Stratum	calls (test)		τ=3	τ=7	τ=15	τ=30
All	514 (95)	real	0.435	0.328	0.171	0.220
		global shuffle	0.454	0.327	0.141	0.206
		increment	**−0.019**	**0.001**	**0.030**	**0.014**
64 kbps	281 (39)	real	0.370	0.250	0.109	0.310
		global shuffle	0.383	0.266	0.133	0.303
		increment	**−0.013**	**−0.016**	**−0.024**	**0.007**
≤24 kbps	232 (55)	real	0.343	0.328	0.174	0.145
		global shuffle	0.443	0.354	0.174	0.128
		increment	**−0.100**	**−0.026**	**−0.001**	**0.018**

Table 14: **Audio on Earnings25 by bitrate stratum.** R^2_{OOS} vs. persistence on Δv for the ridge on WavLM plus past volatility, ticker-disjoint test. With one call per company the same-company swap is the identity, so the random swap is the check. Call counts are inventory calls with audio at that bitrate (one call at 32 kbps sits in neither stratum). The heads see the 475 admitted, priced calls, and the test count is their ticker-disjoint segment. The audio increment never exceeds 0.03, is negative in 7 of the 12 cells, and has no consistent sign across strata. The 64 kbps stratum is not better than the ≤24 kbps one. The full grid (eGeMAPS, emotion2vec+, fusion, MLP heads, all targets) is in `result_table_3_earnings25.csv`.

Head	seg.	τ=3	τ=7	τ=15	τ=30
HAR-RV	in	$0.44 \to 0.46$	$0.37 \to 0.41$	$0.14 \to 0.21$	$-0.29 \to -0.28$
	post	$0.60 \to 0.56$	$0.54 \to 0.51$	$0.35 \to 0.33$	$0.08 \to -0.12$
text + past-vol	in	$0.33 \to -1.71$	$0.28 \to -3.12$	$0.08 \to -3.62$	$-0.08 \to -0.47$
	post	$0.31 \to 0.28$	$0.33 \to 0.27$	$0.23 \to 0.15$	$0.12 \to -0.09$
WavLM + past-vol	in	$0.36 \to 0.26$	$0.26 \to -1.02$	$-0.04 \to -2.01$	$0.05 \to -0.56$
	post	$0.50 \to 0.43$	$0.43 \to 0.32$	$0.15 \to 0.06$	$0.18 \to 0.05$
fusion + past-vol	in	$0.29 \to -1.73$	$0.22 \to -3.13$	$0.04 \to -0.34$	$-0.11 \to -0.45$
	post	$-0.83 \to -9.97$	$0.32 \to -2.29$	$0.19 \to 0.02$	$0.01 \to -0.28$

Table 15: **Anchor sensitivity.** The same models, refit and scored with day 0 on the call date, the anchor of the released label sets (Δv, R^2_{OOS} vs. persistence). Each cell reads measured $\to$ call-date. The ordering on level-v is in the CSV.

Paper	venue	probe (date)	verdict	reason
Qin & Yang, What You Say and How You Say It Matters	ACL 2019	200, 0 code files, pushed 2022-08-04 (2026-10-09)	partial	dataset released (Drive, 5-part zip); no model code in the repository
Yang et al., HTML	WWW 2020	200, 15 code files, pushed 2024-01-05 (2026-10-09)	partial	model classes run verbatim; the released driver imports TensorFlow-1 set_random_seed and torchtext and splits features and labels with independent unseeded shuffles; inputs (WWM-BERT .npy, Praat features) behind dead Drive links, rebuilt; run with a corrected driver
Sawhney et al., multimodal financial forecasting	ACM MM 2020	200, 112 code files, pushed 2020-10-26 (2026-10-09)	partial	TF 2.1 / Keras 2.3.1 / tensorflow-addons 0.8.3 pins; unshipped call inventory, price files and two lexicons; text-feature script returns after one record; no multi-task loss in the released code; ported line by line to PyTorch
Sawhney et al., VolTAGE	EMNLP 2020	200, 5 code files, pushed 2021-11-16 (2026-10-09)	partial	ships the EC split/label files (byte-identical to KeFVP's) and GCN code; embeddings as .pkl; not ported
Niu et al., KeFVP	EMNLP Findings 2023	200, 42 code files, pushed 2024-01-28 (2026-10-09)	partial	requirements.txt does not install as written; the script starts only after five release-defect patches (missing set_seed, package init importing undefined names, a dropped class header, an unreleased 'latent' package, plotting imports); MAEC embeddings never released (regenerated with the authors' generator); EC KePt pickle Drive-only
DialogueGAT	Findings of EMNLP 2022	200, 6 code files, pushed 2023-06-12 (2026-10-09)	partial	MIT code, tau<=15 only; training pickle, SeekingAlpha corpus and CRSP labels never released; DGL has no wheels for torch 2.11; ported (PyG) onto FinCall turns
Yu, Liu & He, Same Company, Same Signal	Findings of ACL 2025	200, 38 code files, pushed 2026-08-13 (2026-10-09)	runnable	MIT; ships DEC with features and rolling masks; PEV/STPEV to 3 decimals, TSMixer 76 cells to 1e-5, TMLP 228 cells within single-seed noise on the authors' embeddings
Yang et al., FinTrust	ACL 2023 short	200, 4 code files, pushed 2024-01-22 (2026-10-09)	partial	perturbation harness + 3/7/15/30 data; complementary control axis, not a volatility model
Sound of Risk	2025	200, 0 code files, pushed 2025-08-26 (2026-10-09)	none	repository holds figures only, zero source files; proprietary 1,795-call corpus
DeFVP	2024	409 (2026-10-09)	none	repository exists but is empty (size 0 KB, never pushed; API returns 409)
ECHO-GL	2024	200, 3 code files, pushed 2024-02-26 (2026-10-09)	none	README states the code 'can not run at present'; target is direction, not volatility
ECC Analyzer	2024	no_url	none	no code released; GPT-4 Turbo + OpenAI embeddings
RiskLabs	2024	no_url	none	no code released; GPT-4/3.5
NumHTML	AAAI 2022	no_url	none	paper promises code; no URL given
GNA-Vol		no_url	none	artefact URL returns 404 (checked 2026-08-23)
AMA-LSTM		no_url	none	10-byte stub repository, unattributed
AT-FinGPT		no_url	none	paywalled; no artefact

Table 16: **Code-availability audit of the ECC volatility literature** (regenerated by `ecvol audit code` with a live GitHub probe on the date shown). One of seventeen papers ships code that runs as released. Seven ship code with a broken environment, missing data, or defects that must be repaired before it starts (HTML's driver imports TensorFlow-1 symbols and decorrelates features from labels, and KeFVP needs five release-defect patches). Nine ship nothing usable. The reason column records what the faithful-reproduction sprint (Table 12) did with each.

Report	Why	Provided by
A company-disjoint split next to the temporal one	On the time-ordered splits in use, 44–80% of test calls come from companies in training, and a probe reads the company out of a frozen call embedding 89.5% of the time (Tables 1, 3).	ticker-disjoint splits for FinCall-Surprise and MAEC (0.0% overlap). The audit command reports the share of test companies seen in training and scores seen and unseen companies separately.
A gap of at least the longest horizon between training and test	Five of six shipped boundaries have a zero-day gap while targets reach 30 days past the call, so training labels contain test-period outcomes.	Released temporal splits with a 30-session gap. The audit command reports the gap in business days and counts training windows that reach into the test period.
Past-volatility baselines (persistence, HAR-RV, and a constant on change targets)	They are hard to beat and often missing from published comparisons. On the measured anchor the constant already scores 0.04–0.20 on Δv (Table 2).	Stage 0 of the harness and the floors in the audit command, including a company-only model.
Same-company and random swaps	Where companies recur in training, a same-company swap moves the text model by less than 0.03 while a random swap hurts, which attributes the score to the company rather than the call (Table 3).	`ecvol controls` for feature-level swaps. Prediction-side swaps in the audit command for any prediction file.
Day 0 measured from the release time	Under the call-date anchor, day 0 moves for 53–72% of calls and the reaction session sits in the pre window (§3).	Release times from 8-K filings in the timing tables. Call-date targets shipped as a labelled variant for comparison with prior work.
The horizon definition, calendar or trading days	The two definitions share only 0.78 correlation at $\tau=3$ and the persistence MSE is 44–63% higher under calendar days, so an MSE without its definition cannot be compared.	Targets under both definitions in every archive and a convention column in every result table.

Table 17: **The reporting checklist of §7.** Each item with the evidence behind it and the part of the release that provides it.